- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

14 In this question the **i** and **j** vectors are horizontal and vertically upward respectively.

A particle of mass 5 kg is at rest on a rough horizontal shelf. The coefficient of friction between the particle and the shelf is μ.

(a) A force $\mathbf{P} = 9\mathbf{i} + 20\mathbf{j}$ N acts on the particle. The particle is on the point of sliding along the shelf.

Determine the value of μ. **[5]**

The force **P** is removed. One end of the shelf is lifted so that it is inclined at $\alpha°$ to the horizontal.

(b) The particle is on the point of sliding **down** the shelf.

Show that $\alpha = 17.2$ to **3** significant figures. **[4]**

(c) The particle is projected **up** the shelf with an initial speed of $5\,\text{m s}^{-1}$.

Given that the particle remains in contact with the shelf, determine the time after projection at which the particle first comes to rest. **[6]**